Learn JavaScript: Beginner's Course

Lesson 1: Introduction & Setup

Introduction & Setup - Study Notes

Key Concepts

• **Integrated Development Environment (IDE)** – VS /Code as a lightweight, extensible code editor.

• **Browser Console** – The developer tools console where JavaScript can be executed instantly.

• **Hello World** – The classic first program that outputs a simple message to verify the environment works.

• **File Extensions & Execution Context** – `.js` files for Node/VS /Code vs. inline `<script>` or console snippets for the browser.

• **Basic Syntax** – `console.log()` statement, semicolons, and whitespace rules.

Summary

Getting started with JavaScript requires only a few tools: a code editor (VS /Code) and a web browser. VS /Code provides syntax highlighting, IntelliSense, and debugging capabilities, making it easier to write and manage `.js` files. The browser's developer console offers an immediate sandbox where you can type JavaScript expressions and see results without creating a file. The traditional "Hello World" program —using `console.log("Hello World");`—serves as a sanity check that both the editor and the console are correctly set up. By installing VS /Code, opening a browser console, and running the Hello World snippet, beginners confirm that their development environment is functional before moving on to more complex concepts like variables, functions, and DOM manipulation.

Important Points

• **Install VS /Code**: Download from `code.visualstudio.com`, run the installer, and optionally add the "Open with Code" context menu.

• **Extensions**: Consider installing *ESLint*, *Prettier*, and *Debugger for Chrome* to enhance the JS workflow.

• **Browser Console**: Open with `F12` or `Ctrl+Shift+I` (Windows/Linux) / `Cmd+Option+I` (macOS), then select the **Console** tab.

• **Hello World in VS /Code**: Create a file `hello.js`, type `console.log("Hello World");`, save, and run via the integrated terminal (`node hello.js`) or the **Run Code** button (if Code Runner extension is installed).

• **Hello World in Browser Console**: Type `console.log("Hello World");` directly and press **Enter**; the string appears in the console output.

• **Syntax Tips**: JavaScript statements usually end with a semicolon; whitespace is ignored, but consistent indentation improves readability.

• **Debugging**: Use `console.log()` to inspect values; later learn to set breakpoints in VS /Code's debugger.

Examples

Example 1 – Hello World in VS /Code (Node)

```javascript
// hello.js
console.log("Hello World");
```

Explanation

1. Create a new file named `hello.js` in VS /Code.
2. Type the line above; the `console.log` function writes the string to the standard output.
3. Open the integrated terminal (`Ctrl+``) and run `node hello.js`.
4. You should see `Hello World` printed in the terminal, confirming that Node.js and VS /Code are correctly configured.

Example 2 – Hello World in Browser Console

1. Open any webpage (e.g., `about:blank`) and press `F12` !’ **Console**.
2. Type the following and hit **Enter**:
   ```javascript
   console.log("Hello World");
   ```
3. The console will display:
   ```
   Hello World
   undefined
   ```
   (`undefined` is the return value of `console.log`.)

Explanation

This shows that the browser’s JavaScript engine can execute code instantly, making it ideal for quick tests and learning snippets.

Quick Review

1. What are two ways to run a “Hello World” JavaScript snippet?
2. Which keyboard shortcut opens the developer console in most browsers?
3. Name one useful VS /Code extension for JavaScript development and its purpose.
4. What does `console.log()` return when called?
5. Why is it a good practice to save JavaScript files with the `.js` extension?

Further Reading

- **JavaScript Basics** – variables, data types, and operators (MDN Web Docs).
- **Working with the DOM** – manipulating HTML elements from JavaScript.
- **Node.js Runtime** – running JavaScript outside the browser, npm basics.
- **ES6+ Features** – arrow functions, template literals, let/const.
- **Debugging in VS /Code** – setting breakpoints, watch expressions, call stack.

These notes are intended to solidify the foundational steps of installing VS /Code, using the browser console, and writing your first JavaScript program.